

Author’s note: AI was used to check grammar and rephrase for better wording. The prompt used was: “Please check grammar, cohesiveness and correct wording of the following.”

Mission Statement:

A coordinated multi-agent system deployed to Europa’s surface to autonomously characterize ice shell thickness and identify an optimal cryobot insertion site without real-time Earth support. The destination is Europa, chosen because its subsurface ocean makes it one of the most promising locations in the Solar System for astrobiology and future exploration. The mission will deploy a surface rover equipped with a ground-penetrating radar sounder, supported by two orbital reconnaissance agents that provide terrain mapping, communications relay, and cooperative localization. By leveraging radar sounding techniques analogous to the Europa Clipper REASON instrument, the system will generate detailed subsurface ice maps to support future cryobot missions.

Operational Phases:

The mission consists of six main phases: Launch and Commissioning, Cruise, Europa Orbit Insertion and Reconnaissance, Landing, Surface Checkout and Site Characterization, Surface Survey and Ice Characterization, and End-of-Mission Operations. Launch and cruise prepare the system for arrival at Europa. Upon arrival, the orbital agents conduct reconnaissance and identify a suitable landing region. Landing is followed by rover deployment, system checkout, and sensor calibration. Main surface operations focus on autonomous rover traversal while collecting radar measurements of the ice shell and building a subsurface map. The orbital agents continuously update terrain maps and relay communications. The final phase includes science data downlink, candidate cryobot site selection, optional extended operations, and safe system passivation.

Systems Roles:

The rover operates autonomously during critical events such as landing, navigation, radar sounding, and site-selection activities. The two orbital agents provide terrain reconnaissance, communications relay, cooperative localization support, and validation of shared mapping data. Ground teams remain in the loop during cruise, strategic mission planning, and anomaly resolution, but all tactical decisions must be performed onboard due to the approximately 43–44 minute one-way communication delay between Earth and Jupiter. Radiation exposure, cryogenic temperatures, communication delays, and limited bandwidth are key operational constraints. The radar sounder provides continuous subsurface measurements that support ice-thickness estimation and cryobot site identification throughout the mission.

References

References

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- [2] NASA, “Radar for Europa Assessment and Sounding: Ocean to Near-surface (REASON),” Europa Clipper Mission. [Online]. Available: <https://europa.nasa.gov/spacecraft/instruments/reason/>. [Accessed: Jun. 21, 2026].
- [3] National Academies of Sciences, Engineering, and Medicine, *Origins, Worlds, and Life: A Decadal Strategy for Planetary Science and Astrobiology 2023–2032*. Washington, DC: National Academies Press, 2022.